Spatiotemporal Risk-Aware Graph Attention Network for Financial Fraud Detection: FFD-STRA-GAT

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I. Introduction

Financial fraud detection in large-scale transaction networks requires robust modeling of node interactions, temporal dynamics, and anomaly propagation. The proposed spatiotemporal attentional based framework integrates graph-based representations with deviation-driven anomaly scoring.

II. Graph Representation

Let $G = (V, E)$ be a directed transaction graph, where V is the vertex set and E is the edge set.

A. Adjacency Matrix

Since this is an adjacency matrix, elements can either be 0 or 1, with
$0 \rightarrow$ no transaction
$1 \rightarrow$ transaction.

$$A_{ij} \in \{0,1\}, \quad A_{ii} = \begin{cases} 1, & \text{if self-transaction exists} \\ 0, & \text{otherwise} \end{cases}$$

A forced complete graph yields complexity: $O(n^2)$

This is a hypothetical situation where every account in the data set is connected to other accounts, and will most likely never occur in real life(but it is safer to include a rule like this).

A sparse structure reduces it to: $O(kn)$

III. Feature Representation

To represent the data set, the number of features is 17 in number(inclusive of sender node, receiver node and edge features), as:

$$X = \begin{bmatrix} H_i \\ H_j \\ \text{Age}_i \\ \text{Age}_j \\ \text{Loc}_i \\ \text{Loc}_j \\ T_{\text{avg},i} \\ T_{\text{avg},j} \\ T_{\text{avg},ij} \\ \epsilon_i \\ \epsilon_j \\ \epsilon_{ij} \\ \text{Cat}_i \\ \text{Cat}_j \\ F_{ij} \\ R_{ij} \\ \Delta_{ij} \end{bmatrix}$$

essentially a 17x1 feature vector.

A. Node Features

Sender: $X_i = (H_i, Age_i, Loc_i, T_{avg,i}, \epsilon_i, Cat_i)$
Receiver: $X_j = (H_j, Age_j, Loc_j, T_{avg,j}, \epsilon_j, Cat_j)$

B. Edge Features

$e_{ij}^{(t)} = (F_{ij}^{(t)}, \Delta_{ij}^{(t)}, R_{ij}^{(t)}, \epsilon_{ij})$

C. Transaction Deviation

This feature is very important during the calculation of Anomaly Score, $AS_{ij}^{(t)}$

$$\Delta_{ij}^{(t)} = |F_{ij}^{(t)} - T_{avg,ij}|$$

D. Average Transaction Value

$$T_{avg,ij} = \frac{\sum_{t=1}^{n-1} F_{ij}^{(t)}}{n-1}$$

The temporal update of the average transaction value is, then, given by:

$$T_{avg,ij}^{(t+1)} = \frac{n_{ij}^{(t)} T_{avg,ij}^{(t)} + F_{ij}^{(t+1)}}{n_{ij}^{(t)} + 1}$$

E. Standard Deviation-temporal update

$$\sigma_{ij}^{(t)} = \sqrt{\frac{1}{n_{ij}^{(t)}} \sum_t (F_{ij}^{(t)} - T_{avg,ij}^{(t)})^2}$$

F. Category Mismatch

$$Cat_{mis} = \begin{cases} 1, & \text{if } Cat_i \neq Cat_j \\ \gamma, & \text{if } Cat_i = Cat_j \end{cases}$$

Here, γ should be learnable, but for the sake of keeping this paper a little simple, I have used $\gamma = 0.5$. However, I plan on make γ learnable in my next update of the document.

G. Relationship Strength

$$R_{ij}^{(t)} = \frac{n_{ij}^{(t)}}{\sum_k n_{ik}}$$

Update:

$$n_{ij}^{(t+1)} = n_{ij}^{(t)} + 1$$

This is when the relationship is static, not learnable

H. Tolerance Model

$$\epsilon_{ij} = \alpha\epsilon_i + (1-\alpha)\epsilon_j + \Psi\sigma_{ij}$$

I. Unweighted Minimal Anomaly Score

$$AS_{ij}^{(t)} = \frac{\Delta_{ij}^{(t)} Cat_{mis}}{\epsilon_{ij}}$$

J. Temporal Weight Updation-Static

$$w_{ij}^{(t+1)} = w_{ij}^{(t)} \cdot \log\left(c + \exp\left(\frac{\Delta_{ij}^{(t)} Cat_{mis}}{\epsilon_{ij}}\right)\right)(\pi + \lambda R_{ij}^{(t)})$$

K. Node-Level Aggregation

$$\phi_{ij}^{(t)} = w_{ij}^{(t)} \cdot AS_{ij}^{(t)}$$

$$\rho_i^{(t)} = \log\left(\sum_{j \in \mathcal{N}(i)} \exp(\phi_{ij}^{(t)})\right)$$

IV. Piecewise Weighted Anomaly Function

$$AS_i^{(W,t)} = \begin{cases} \rho_i^{(t)}, & \rho_i^{(t)} \leq \Theta_i^{(t)} \\ \max_j \phi_{ij}^{(t)}, & \text{otherwise} \end{cases}$$

V. Threshold Definition-static

$$\Theta_i^{(t)} = Q_{1-\alpha}(\rho_i^{(t)}) + \log(|\mathcal{N}(i)|)$$

VI. Fraud Criterion

$$AS_i^{(W,t)} > \Theta_i^{(t)} \Rightarrow \text{Fraud}$$

VII. Learnable GAT Extension

A. Node Transformation

$$h_i = W^{(t)} X_i$$

B. Attention Score

$$Ats_{ij}^{(t)} = LeakyReLU(a^T \tilde{z}_{ij})$$

where, $\tilde{z}_{ij}^{(t)} = [W^{(t)} X_i, W^{(t)} X_j, e_{ij}^{(t)}]^T$
Moreover,
$dim(W^{(t)} X_i) = dim(W^{(t)} X_j) = d_h$, and
$dim(e_{ij}^{(t)}) = d_e$
$a^T \rightarrow$ attention vector

1) Construction of learnable attention vector, a^T: We initialize, $a_0 \in$ U$(-\epsilon, \epsilon)$
where ϵ=$\sqrt{\frac{6}{d_z}}$, where d_z=$dim(z_{ij}^{(t)})$, and
$d_z = 2d_h + d_e$
$a_{k+1} \leftarrow a_k - \eta\frac{\partial L}{\partial a_k}$

a$^T = \sum_{k=1}^{d_z} a_k \delta_k^T$

δ_k^T $\rightarrow$row basis vector

C. Learnable Tolerance Updation

$$\epsilon_{ij}^{(t)} = \alpha_{ij}^{(t)}\epsilon_i + (1-\alpha_{ij}^{(t)})\epsilon_j + \Psi_{ij}^{(t)}\sigma_{ij}$$

Now, to make $\Psi_{ij}^{(t)}$ we do the following:
$d^T = \sum_{k=1}^{d_z} d_k \delta_k^T$

We initialize, $d_0 \in$ U$(-\epsilon, \epsilon)$

where ϵ=$\sqrt{\frac{6}{d_z}}$ (I have used Xavier updation), where d_z=$dim(z_{ij}^{(t)})$, and
$d_z = 2d_h + d_e$
$d_{k+1} \leftarrow d_k - \eta\frac{\partial L}{\partial d_k}$
$\Psi_{ij}^{(t)} = Softplus(d^T \tilde{z}_{ij}^{(t)})$

Now we make $\alpha_{ij}^{(t)}$ learnable:
$b^T = \sum_{k=1}^{d_z} b_k \delta_k^T$

We initialize, $b_0 \in$ U$(-\epsilon, \epsilon)$

where ϵ=$\sqrt{\frac{6}{d_z}}$ (I have used Xavier updation),
where d_z=$dim(z_{ij}^{(t)})$, and
$d_z = 2d_h + d_e$
$b_{k+1} \leftarrow b_k - \eta\frac{\partial L}{\partial b_k}$
$\alpha_{ij}^{(t)} = \sigma(b^T \tilde{z}_{ij}^{(t)})$

D. Learnable Unweighted Minimal Anomaly Score

Overriding the previous definition of the Unweighted Minimal Anomaly Score, we now define it by:
$AS_{ij}^{(t)} = \frac{\Delta_{ij}^{(t)} Cat_{mis}}{\epsilon_{ij}^{(t)}}$ approximates a scale-invariant anomaly

intensity, analogous to a z-score but with a learnable tolerance replacing variance.
Properties of $AS_{ij}^{(t)}$:
(a)$0 \leq AS_{ij}^{(t)}$
(b)$\Delta_{ij}^{(t)} \uparrow$, $AS_{ij}^{(t)} \uparrow$
(c)$AS_{ij}^{(t)}$ is scale invariant.
$AS_{ij}^{(t)}$ is not a probability, but a normalized deviation surrogate

E. Attention Normalization of Relationship Strength

$$R_{ij}^{(t)} = \frac{\exp(Ats_{ij}^{(t)})}{\sum_{k \in \mathcal{N}(i)} \exp(Ats_{ik}^{(t)})}$$

F. Learnable Weight Updation

$$w_{ij}^{(t+1)} = w_{ij}^{(t)} \cdot \log\left(c + \exp\left(\frac{\Delta_{ij}^{(t)} Cat_{mis}}{\epsilon_{ij}}\right)\right)(\pi + \lambda_{ij}^{(t)} R_{ij}^{(t)})$$

We fix $\pi = 1$, and $c = 1$. The function is a smooth, bounded-growth amplification function inspired by $Softplus(x)$.
Now, to make $\lambda_{ij}^{(t)}$ we do the following:
$c^T = \sum_{k=1}^{d_z} c_k \delta_k^T$
We initialize, $c_0 \in \mathrm{U}(-\epsilon, \epsilon)$
where ϵ=$\sqrt{\frac{6}{d_z}}$ (I have used Xavier updation), where d_z=$dim(z_{ij}^{(t)})$, and
$d_z = 2d_h + d_e$
$c_{k+1} \leftarrow c_k - \eta \frac{\partial L}{\partial c_k}$
$\lambda_{ij}^{(t)} = Softplus(c^T \tilde{z}_{ij}^{(t)})$

G. Aggregation

$$\rho_i^{(t)} = \rho_i^{(t)}(R_{ij}^{(t)}, h_j, e_{ij})$$

Implemented via attention-weighted aggregation.

H. Learnable Weighted Anomaly Function

$$AS_i^{(W,t)} = \begin{cases} \rho_i^{(t)}, & \rho_i^{(t)} \leq \Theta_i^{(t)} \\ \max_j \phi_{ij}^{(t)}, & \text{otherwise} \end{cases}$$

where,

$$\phi_{ij}^{(t)} = w_{ij}^{(t)} \cdot AS_{ij}^{(t)}$$

Why did I use max, and not $mean$ or $Softmax$?
This is because max captures worst-case risk propagation. This ensures that once the aggregate crosses the threshold, the function prioritizes the most suspicious transaction thereby enabling sharp localization of fraud sources.

I. Theoretical Interpretation of the Anomaly Score

The unweighted minimal anomaly score is defined as:

$$AS_{ij}^{(t)} = \frac{\Delta_{ij}^{(t)} Cat_{mis}}{\epsilon_{ij}^{(t)}} = \frac{\left|F_{ij}^{(t)} - T_{avg,ij}^{(t)}\right| Cat_{mis}}{\epsilon_{ij}^{(t)}} \quad (1)$$

where $F_{ij}^{(t)}$ is the current transaction value, $T_{avg,ij}^{(t)}$ is the historical mean, and $\epsilon_{ij}^{(t)}$ is a learnable tolerance parameter.

The proposed anomaly score can be interpreted as a *generalized z-score*, where the standard deviation is replaced by a context-aware, learnable scale parameter. Unlike the classical z-score,

$$Z_{ij}^{(t)} = \frac{(F_{ij}^{(t)} - T_{avg,ij}^{(t)}) Cat_{mis}}{\sigma_{ij}^{(t)}}, \quad (2)$$

the formulation of $AS_{ij}^{(t)}$ allows adaptive sensitivity across nodes and edges via:

$$\epsilon_{ij}^{(t)} = \alpha_{ij}^{(t)} \epsilon_i + (1 - \alpha_{ij}^{(t)}) \epsilon_j + \Psi_{ij}^{(t)} \sigma_{ij}^{(t)}. \quad (3)$$

The anomaly score satisfies the following properties:
(i) non-negativity, $AS_{ij}^{(t)} \geq 0$;
(ii) monotonicity, $\Delta_{ij}^{(t)} \uparrow \Rightarrow AS_{ij}^{(t)} \uparrow$;and
(iii) scale invariance under proportional scaling of $\epsilon_{ij}^{(t)}$.
Thus, $AS_{ij}^{(t)}$ quantifies the relative intensity of deviation from expected behavior.

Importantly, $AS_{ij}^{(t)}$ is not a probability measure but a normalized deviation surrogate, enabling flexibility under non-Gaussian and heterogeneous transaction patterns. By incorporating learnable tolerance and graph-aware adaptation, the proposed formulation extends classical statistical deviation measures to a spatiotemporal, graph-based anomaly detection setting.

VIII. Learnable Threshold Definition

Discarding the previously written static definition of $\Theta_i^{(t)}$, we will use the following the definition:

$$\Theta_i^{(t)} = \mu_i^{(t)} + k_i^{(t)} . \sigma_i^{(t)} + \log(|\mathcal{N}(i)|)$$

where, $\mu_i^{(t)}, k_i^{(t)}$, and $\sigma_i^{(t)}$ are the learnable mean, sensitivity parameter and standard deviation respectively.

$$\mu_i^{(t)} = [1 - \beta_i^{(t)}]\mu_i^{(t-1)} + \beta_i^{(t)} AS_i^{(W,t)}$$

where,

$$\beta \rightarrow stability$$

$$\sigma_i^{2,(t)} = [1 - \beta_i^{(t)}]\sigma_i^{2,(t-1)} + \beta_i^{(t)}[AS_i^{(W,t)} - \mu_i^{(t)}]^2$$

$$\beta_i^{(t)} = \sigma([u^{T,(t)} * h_i^{(t)}] + b_{\tilde{\beta}})$$

Here, $b_{\tilde{\beta}}$ is the bias that acts as a scalar offset to control initial smoothing.

$$\tilde{\beta}^{(0)} \in [0.02, 0.08]$$

$$b_{\tilde{\beta}}^{(t)} = \log\left(\frac{\tilde{\beta}^{(t-1)}}{1 - \tilde{\beta}^{(t-1)}}\right) - u^{T,(t)} h_i^{(t)}$$

$$\tilde{\beta}^{(t)} = \sigma(\tilde{\beta}^{(t-1)} + \eta_{\tilde{\beta}}^{(t)}(AS_i^{(W,t)} - \mu_i^{(t)}))$$

$$\eta_{\tilde{\beta}}^{(t)} = \eta_0 \sigma(AS_i^{(W,t)} - \mu_i^{(t)})$$

where, $\eta_0 \in$[0.03,0.07]

Now we aim to define both $u^{T,(t)}$ and $k_i^{(t)}$, and make them learnable:

$u^{T,(t)} = \sum_{k=1}^{d_z} u_k \delta_k^T$

We initialize, $u_0 \in$ U$(-\epsilon, \epsilon)$
where ϵ=$\sqrt{\frac{6}{d_z}}$ (I have used Xavier updation),
where d_z=$dim(z_{ij}^{(t)})$, and
$d_z = 2d_h + d_e$
$u_{k+1} \leftarrow u_k - \eta \frac{\partial L}{\partial u_k}$

$v^T = \sum_{k=1}^{d_z} v_k \delta_k^T$

We initialize, $v_0 \in$ U$(-\epsilon, \epsilon)$
where ϵ=$\sqrt{\frac{6}{d_z}}$ (I have used Xavier updation),
where d_z=$dim(z_{ij}^{(t)})$, and
$d_z = 2d_h + d_e$
$v_{k+1} \leftarrow v_k - \eta \frac{\partial L}{\partial v_k}$
$k_i^{(t)} = Softplus(v^T h_i^{(t)} + b_k^{(t)})$

Definition of $b_k^{(t)}$:

We initialize, $b_k^{(0)} \in$ U$(-\epsilon, \epsilon)$
where ϵ=$\sqrt{\frac{6}{d_z}}$ (I have used Xavier updation),
where d_z=$dim(z_{ij}^{(t)})$, and
$d_z = 2d_h + d_e$
$b_{k+1}^{(t)} \leftarrow b_k^{(t)} - \eta \frac{\partial L}{\partial b_k^{(t)}}$
$b_k^{(t+1)} = b_k^{(t)} - \eta \sum_i \frac{\partial L}{\partial k_i^{(t)}} [\sigma(v^T h_i^{(t)} + b_k^{(t)})]$

IX. Output Layer

$$f_i^{(t)} = \frac{\Theta_i^{(t)} - AS_i^{(W,t)}}{\sqrt{\sigma_i^{(t)}} + 1}$$

$$p_i^{(t)} = \sigma(f_i^{(t)})$$

X. Loss Function

$$\mathcal{L} = -\sum_i \left[\tau y_i \log |p_i^{(t)}| + (1 - y_i) \log |1 - p_i^{(t)}| \right] + \nu \|\Omega\|^2;$$

$$\tau > 1, \nu > 0$$

XI. Forward Propagation

Step 1: Node Embedding

$$h_i = W^{(t)} X_i$$

Step 2: Attention Score

$$Ats_{ij}^{(t)} = a^T \tilde{z}_{ij}$$

Step 3: Normalization

$$R_{ij}^{(t)} = \frac{\exp(Ats_{ij}^{(t)})}{\sum_{k \in \mathcal{N}(i)} \exp(Ats_{ik}^{(t)})}$$

Step 4: Aggregation

$$\rho_i^{(t)} = \rho_i^{(t)}(R_{ij}^{(t)}, h_j, e_{ij})$$

Step 5: Output

$$f_i^{(t)} = \frac{\Theta_i^{(t)} - AS_i^{(W,t)}}{\sigma_i^{(t)} + 1}$$

$$p_i^{(t)} = \sigma(f_i^{(t)})$$

XII. Backward Propagation

$$\frac{\partial \mathcal{L}}{\partial \Omega} = \frac{\partial \mathcal{L}}{\partial p_i^{(t)}} \cdot \frac{\partial p_i^{(t)}}{\partial f_i^{(t)}} \cdot \frac{\partial f_i^{(t)}}{\partial AS_i^{(W,t)}} \cdot \frac{\partial AS_i^{(W,t)}}{\partial \rho_i^{(t)}} \cdot \frac{\partial \rho_i^{(t)}}{\partial R_{ij}^{(t)}} \cdot \frac{\partial R_{ij}^{(t)}}{\partial Ats_{ij}^{(t)}} \cdot \frac{\partial Ats_{ij}^{(t)}}{\partial \Omega}$$

XIII. Algorithmic Framework(Page:5)

XIV. Complexity Analysis

Let $n = |V|, m = |E|, d$ be embedding dimension, P workers.

A. Time Complexity

$$O\left(\frac{n}{P} d^2\right) + O\left(\frac{m}{P} d\right) + O\left(\frac{n+m}{P} d\right)$$

$$= O\left(\frac{n+m}{P} d\right)$$

B. Communication Complexity

$$O(m_b d) + O(|\Omega| \log P)$$

C. Space Complexity

$$O(|\Omega| + \frac{n+m}{P} d)$$

XV. Mini-Batch Extension

Let batch size be B, and neighbors be k.

$$Complexity = O(Bkd)$$

Algorithm 1 FFD-STRA-GAT

Require: $G = (V, E), X_j, X_i, e_{ij}, y_i$
Ensure: $p_i^{(t)}$

Initialize $\Omega = \{W^{(t)}, a^T, b^T, c^T, d^T, v^T\}$
repeat
  for $i \in V$ **do**
    $h_i \leftarrow W^{(t)} X_i$
  end for
  for $(i, j) \in E$ **do**
    $Ats_{ij}^{(t)} \leftarrow a^T \tilde{z}_{ij}$
  end for
  for $i \in V$ **do**
    Compute $R_{ij}^{(t)}$
  end for
  for $i \in V$ **do**
    $\rho_i^{(t)} \leftarrow \rho_i^{(t)}(R_{ij}^{(t)}, h_j, e_{ij})$
  end for
  for $i \in V$ **do**
    Compute $AS_i^{(W,t)}$
    $f_i^{(t)} \leftarrow \frac{\Theta_i^{(t)} - AS_i^{(W,t)}}{\sigma_i^{(t)} + 1}$
    $p_i^{(t)} \leftarrow \sigma(f_i^{(t)})$
  end for
  Compute $\mathcal{L}$
  $\Omega \leftarrow \Omega - \eta \nabla_\Omega \mathcal{L}$
until convergence

XVI. Scalability Theorem

Theorem: If $m_b \ll m$, then near-linear speedup holds.
Proof Sketch:

$$T_{comp} = O\left(\frac{n+m}{P} d\right)$$

$$T_{comm} = O(m_b d + \|\Omega\| \log P)$$

If $T_{comm} \ll T_{comp}$:

$$T \approx O\left(\frac{n+m}{P} d\right)$$

XVII. Illustrative Graph Representation(Page:5)

XVIII. Distributed Architecture (DDP)(Page:6)

XIX. Model Pipeline(Page:6)

XX. Anomaly Propagation Illustration(Page:6)

XXI. GNN Architecture(Page:7)

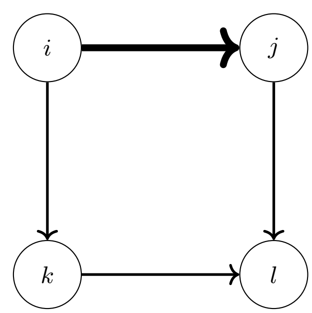
XXII. Future updates

I aim to make γ(in the definition of Category mismatch), τ and ν(both from loss function), learnable

Algorithm 2 Distributed model using DDP

Require: Partitioned $G = (V, E)$ across P workers
Ensure: $p_i^{(t)}$

Initialize Ω
repeat
  Parallel on workers
  for $i \in V_p$ **do**
    $h_i \leftarrow W^{(t)} X_i$
  end for
  for $(i, j) \in E_p$ **do**
    $Ats_{ij}^{(t)} \leftarrow a^T \tilde{z}_{ij}$
  end for
  Exchange boundary $Ats_{ij}^{(t)}$
  for $i \in V_p$ **do**
    Compute $R_{ij}^{(t)}$
    $\rho_i^{(t)}$
  end for
  for $i \in V_p$ **do**
    Compute $AS_i^{(W,t)}$
    $f_i^{(t)}$
    $p_i^{(t)}$
  end for
  Compute $\mathcal{L}_p$
  All-reduce gradients
  $\Omega \leftarrow \Omega - \eta \nabla_\Omega \mathcal{L}$
until convergence



Edge weight $\propto R_{ij}^{(t)}$, anomaly $\propto AS_{ij}^{(t)}$

Fig. 1. A simple clean graph structure used in the attention framework.

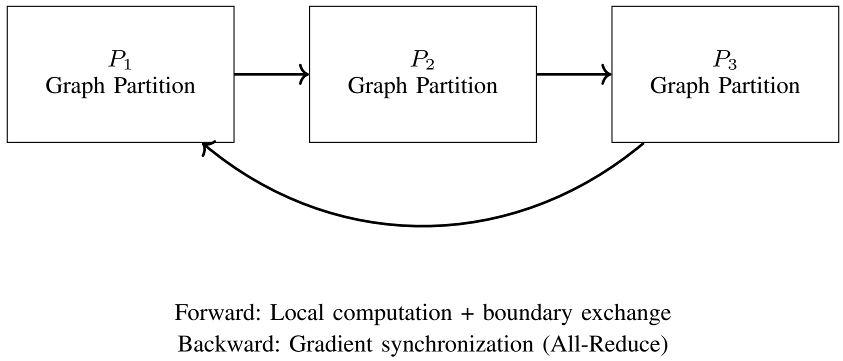


Fig. 2. Distributed Data Parallel training with partitioned graph computation.

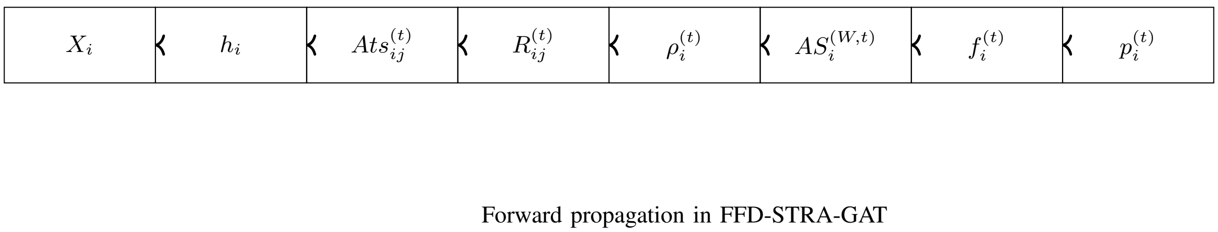


Fig. 3. End-to-end pipeline of the FFD-STRA-GAT framework.

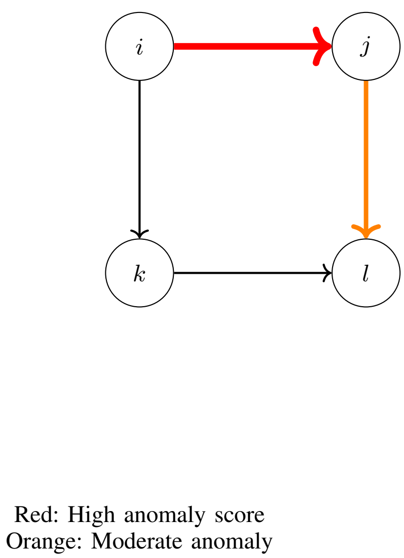
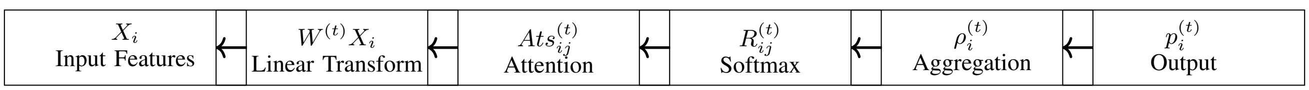


Fig. 4. Anomaly propagation across the graph.



Layer-wise computation in FFD-STRA-GAT

Fig. 5. Layered architecture of the FFD-STRA-GAT model.